

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –**RAIGAD -402 103****Semester Examination – May/June - 2019****Branch: Electronics and Telecommunication Engg.****Sem.:-IV****Subject with Subject Code:-NMCP (BTBSC406)****Marks:60****Date: 24-05-2019****Time:- 3 Hr.****Instructions to the Students**

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

(Marks)**Q.1. Solve the following questions (Any3)**

- a) Round off the number 37.46381 correct upto four Significant figures and find absolute error, relative error and percentage relative error. (4)
- b) Find absolute error if the number $X = 0.00545828$ is
 - a) truncated to 3 decimal digits
 - b) rounded off to 3 decimal digits. (4)
- c) Define Inherent error and truncation error with examples. (4)
- d) Define absolute error in addition operation and absolute error in multiplication operation. (4)

Q.2. Solve the following questions (Any 2)

- 1) If $x \log_{10} x = 1.2$, find real root by Regula-Falsi method. (6)
- 2) If $x^2 - e^{-x} = 0$, find real root by Newton Raphson method. (6)
- 3) Solve by Gauss Seidal Method the system of equations

$$\begin{aligned} 7x + 2y + 4z &= 16 \\ 2x + y + 7z &= 10 \\ 2x + 3y + 5z &= 2 \end{aligned}$$
 (6)

Q.3. Solve the following questions (Any 3)

- 1) Fit the curve of type $y = a + bx^2$ that fits the following data (4)

x	0	1	2	3
y	2	4	10	15

- 2) Find i) $\Delta^n e^x$ ii) Prove that $\Delta = \mu\delta + \frac{1}{2}\delta^2$ (4)
- 3) Evaluate $f(9)$ by using Newton's divided difference formula for (4)

x	5	7	11	13	17
f(x)	150	392	1452	2366	5202

- 4) Find cubic polynomial which takes the following values and find $f(4)$. (4)

x	0	1	2	3
f(x)	1	2	1	10

Q.4. Solve the following questions (Any 2)

- 1) Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using a) Trapezoidal rule and (6)
b) Simpson's 1/3 rd Rule
- 2) If $\frac{dy}{dx} = x + y^2$, $y(0)=1$, then determine $y(0.02)$ and $y(0.04)$ by Euler's modified method. (6)
- 3) If $\frac{dy}{dx} = \sqrt{(x+y)}$, $y(0)=1$, then determine $y(0.2)$ by 4th order Runge-Kutta method. (6)

Q.5. Solve the following questions (Any 3)

- a) Explain input and output operators cin and cout with the help of C++ example (4)
- b) With the help of suitable C++ example explain the concept of inline function. (4)
- c) Write a program to explain the concept of class and object (4)
- d) Explain the concept of function overloading with the help of C++ example (4)

Q.6. Solve the following questions in detail.

- a) What is inheritance? What are the different types of inheritance? Draw the block diagram of each type of inheritance? Explain any one type of inheritance in detail. (6)
- b) Write and explain a program for overloading of unary operator to overload minus(-) operator. (6)